Key features of parabolas

1 Consider the general quadratic equation $y = ax^2 + bx + c$, $a \neq 0$.

a If $a < 0$, in what direction will the parabola open?

b If $a > 0$, in what direction will the parabola open?

2 Does the parabola represented by the equation $y = x^2 - 8x + 9$ open upward or downward?

3 Does the graph of $y = x^2 + 6$ have any x-intercepts? Explain your answer.

4 State whether the following parabolas have x-intercepts:

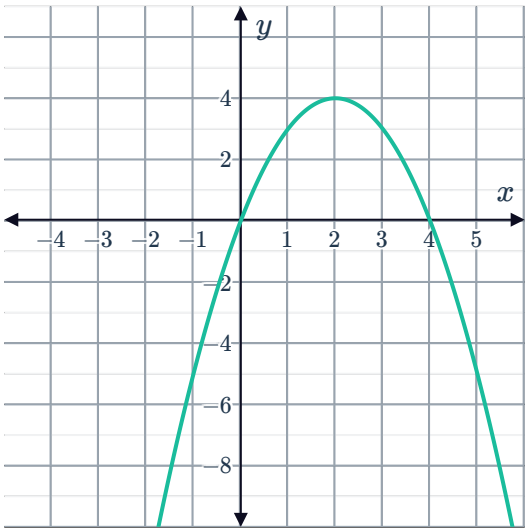
a $y = (x - 7)^2 + 4$

b $y = -(x - 7)^2 + 4$

c $y = -(x - 7)^2 - 4$

d $y = (x - 7)^2 - 4$

5 Consider the given graph:

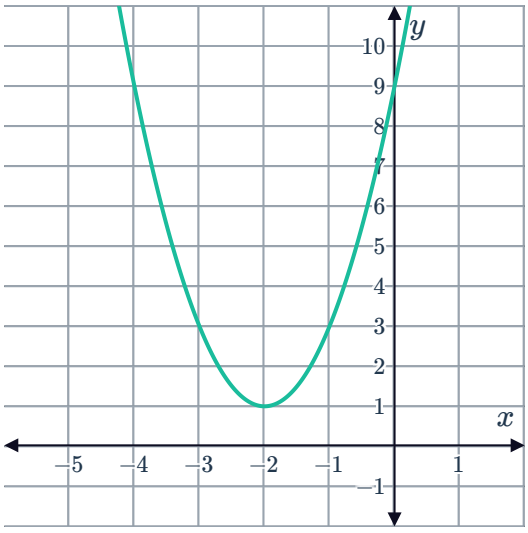


a State the x-intercepts.

b State the y-intercept.

c State the maximum value.

6 Consider the given graph:



a Is the curve concave up or concave down?

b State the y-intercept of the graph.

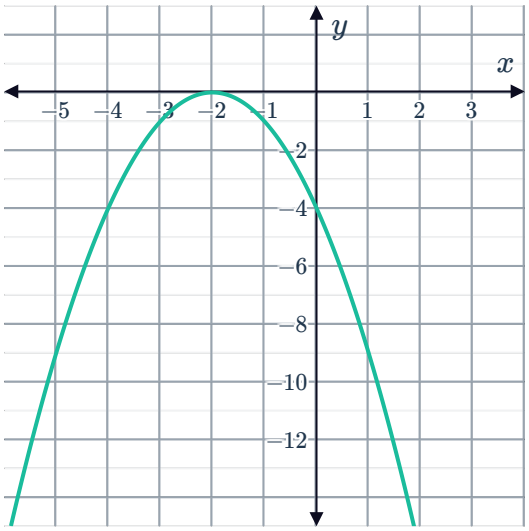
c What is the minimum value?

d At which value of x does the minimum value occur?

e Determine the interval of x for which the graph is decreasing.

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7 Consider the graph of the parabola:



a State the coordinates of the x-intercept.

b State the coordinates of the vertex.

c State whether the following statements are true about the vertex:

i The vertex is the minimum value of the graph.

ii The vertex occurs at the x-intercept.

iii The vertex lies on the axis of symmetry.

iv The vertex is the maximum value of the graph.

8 Suppose that a particular parabola is concave down, and its vertex is located in quadrant 2.

a How many x-intercepts will the parabola have?

b How many y-intercepts will the parabola have?

9 Suppose that a particular parabola has two x-intercepts, and its vertex is located in quadrant 4. Will such a parabola be concave up or concave down?

10 Consider the quadratic function defined in the table on the right:

x	y
-7	11
-6	6
-5	3
-4	2
-3	3
-2	6
-1	11

a What are the coordinates of the vertex?

b What is the minimum value of the function?

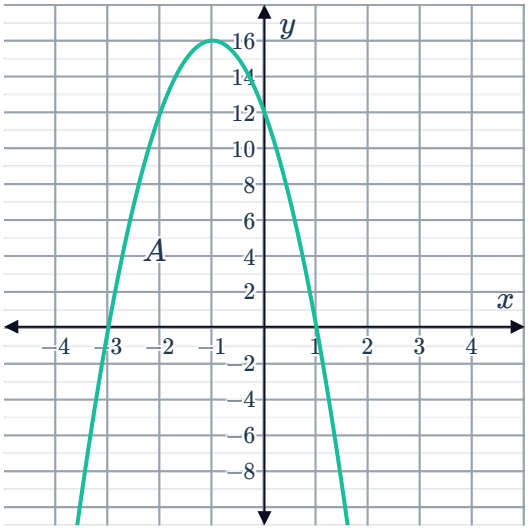
11 A vertical parabola has an x-intercept at $(-1, 0)$ and a vertex at $(1, -6)$. Find the other x-intercept.

12 State whether the following can be found, without any calculation, from the equation of the form $y = (x - h)^2 + k$ but not from the equation of the form $y = x^2 + bx + c$:

a x-intercepts **b** y-intercept **c** vertex

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13 Quadratic function A is represented graphically as shown. Quadratic function B, which is concave down, shares the same x-intercepts as quadratic function A, but has a y-intercept closer to the origin. Which of the functions has a greater maximum value?



14 State the axis of symmetry of the parabola $y = k\,(x - 7)\,(x + 7)$ for any value of k.

15 Consider the equation $y = 25 - (x + 2)^2$. What is the maximum value of y?

16 Consider the function $y = (14 - x)\,(x - 6)$.

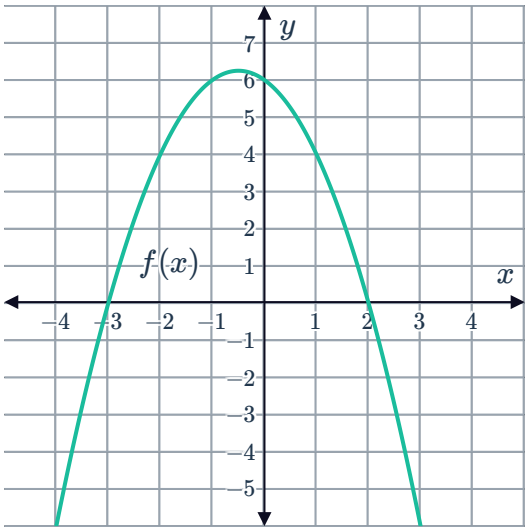
- **a** State the zeros of the function.
- **b** Find the axis of symmetry.
- **c** Is the graph of the function concave up or concave down?
- **d** Determine the maximum y-value of the function.

17 Consider the parabola of the form $y = ax^2 + bx + c$, where $a \neq 0$.
Complete the following statement:
The x-coordinate of the vertex of the parabola occurs at $x = \square$. The y-coordinate of the vertex is found by substituting $x = \square$ into the parabola's equation and evaluating the function at this value of x.

18 Find the x-coordinate of the vertex of the parabola represented by $P\,(x) = px^2 - \frac{1}{2}px - q$.

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19 Consider the graph of the function $f(x) = -x^2 - x + 6$:
Using the graph, write down the solutions to the equation $-x^2 - x + 6 = 0$.



20 True or false:

- **a** The quadratic formula can be used to find the y-intercept.
- **b** If the parabola has only one x-intercept , then the x-intercept is also the vertex.

21 Consider the parabola whose equation is $y = 3x^2 + 3x - 7$. Find the x-intercepts of the parabola in exact form.

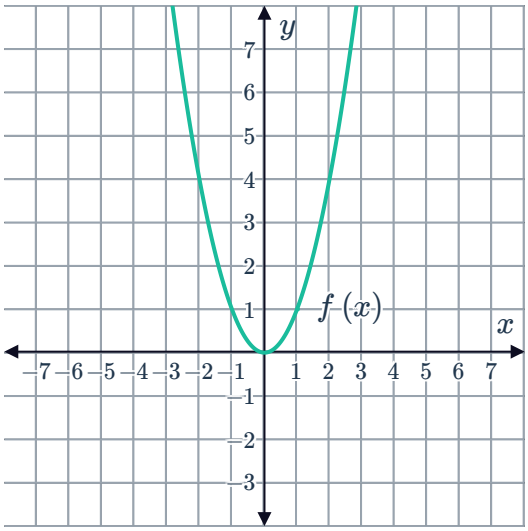
Graphing quadratics

22 Consider the parabola described by the function $y = -2x^2 + 2$.

- **a** Is the parabola concave up or down?
- **b** Is the parabola more or less steep than the parabola $y = x^2$?
- **c** What are the coordinates of the vertex of the parabola?
- **d** Sketch the graph of $y = -2x^2 + 2$.

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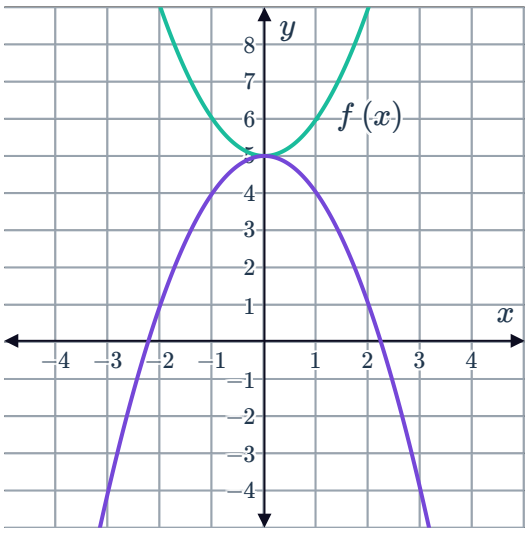
23 A graph of $f(x) = x^2$ is shown on the right. Sketch the graph after it has undergone transformations resulting in the following functions:



a $g(x) = 4x^2 - 2$

b $h(x) = 2(x + 4)^2$

24 Consider the two graphs. One of them has equation $f(x) = x^2 + 5$.
What is the equation of the other graph?



25 Consider the quadratic function $h(x) = x^2 + 2$.

a Sketch the graph of the parabola $h(x)$.

b Plot the axis of symmetry of the parabola on the same graph.

c What is the vertex of the parabola?

26 Consider the parabola described by the function $y = \frac{1}{2}(x - 3)^2$.

a Is the parabola concave up or down?

b Is the parabola more or less steep than the parabola $y = x^2$?

c What are the coordinates of the vertex of the parabola?

d Sketch the graph of $y = \frac{1}{2}(x - 3)^2$.

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27 Consider the equation $y = (x-3)^2 - 1$.

a Find the x-intercepts.

b Find the y-intercept.

c Determine the coordinates of the vertex.

d Sketch the graph.

28 Consider the quadratic function $f(x) = -3(x+2)^2 - 4$.

a What are the coordinates of the vertex of this parabola?

b What is the equation of the axis of symmetry of this parabola?

c What is the y-coordinate of the graph of $f(x)$ at $x = -1$?

d Sketch the graph of the parabola.

e Plot the axis of symmetry of the parabola on the same graph.

29 On a number plane, sketch the shape of a parabola of the form $y = a(x-h)^2 + k$ that has the following signs for a, h and k:

a $a > 0, h > 0, k > 0$

b $a < 0, h > 0, k > 0$

c $a > 0, h > 0, k < 0$

d $a < 0, h > 0, k < 0$

30 Consider the parabola $y = (2-x)(x+4)$.

a State the y-intercept.

b State the x-intercepts.

c Complete the table of values:

x	-5	-3	-1	1	3
y					

d Determine the coordinates of the vertex of the parabola.

e Sketch the graph of the parabola.

31 Consider the parabola $y = (x-3)(x-1)$.

a Find the y-intercept.

b Find the x-intercepts.

c State the equation of the axis of symmetry.

d Find the coordinates of the turning point.

e Sketch the graph of the parabola.

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32 Consider the parabola $y = x\,(x + 6)$.

- **a** Find the y-intercept.
- **b** Find the x-intercepts.
- **c** State the equation of the axis of symmetry.
- **d** Find the coordinates of the turning point.
- **e** Sketch the graph of the parabola.

33 Sketch the graph of the following:

- **a** $y = (x + 2)(x - 3)$
- **b** $y = (x - 3)(x + 1)$

34 Consider the function $y = (x + 5)\,(x + 1)$.

- **a** Sketch the graph.
- **b** Sketch the graph of $y = -\,(x + 5)\,(x + 1)$ on the same set of axes.

35 Consider the equation $y = x^2 - 6x + 8$.

- **a** Factorise the expression $x^2 - 6x + 8$.
- **b** Hence, or otherwise, find the x-intercepts of the quadratic function $y = x^2 - 6x + 8$
- **c** Find the coordinates of the turning point.
- **d** Sketch the graph of the function.

36 Consider the parabola $y = x^2 + x - 12$.

- **a** Factorise the equation of the parabola.
- **b** Find the x-intercepts of the curve.
- **c** Find the y-intercept of the curve.
- **d** What is the equation of the vertical axis of symmetry for the parabola?
- **e** Find the coordinates of the vertex of the parabola.
- **f** Is the parabola concave up or down?
- **g** Sketch the graph of $y = x^2 + x - 12$.

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37 A parabola has the equation $y = x^2 + 4x - 1$.

- **a** Express the equation of the parabola in the form $y = (x - h)^2 + k$ by completing the square.
- **b** Find the y-intercept of the parabola.
- **c** Find the vertex of the parabola.
- **d** Is the parabola concave up or down?
- **e** Hence, sketch the graph of $y = x^2 + 4x - 1$.

38 Consider the quadratic $y = x^2 - 12x + 32$.

- **a** Find the zeros of the quadratic function.
- **b** Express the equation in the form $y = a(x - h)^2 + k$ by completing the square.
- **c** Find the coordinates of the vertex of the parabola.
- **d** Hence, sketch the graph.

39 Consider the curve $y = x^2 + 6x + 4$.

- **a** Determine the axis of symmetry.
- **b** Hence, determine the minimum value of y.
- **c** Sketch the graph of the function.

40 Consider the function $P(x) = -2x^2 - 8x + 2$.

- **a** Find the coordinates of the vertex.
- **b** Sketch the graph.

41 Consider the equation $y = 6x - x^2$.

- **a** Find the x-intercepts of the quadratic function.
- **b** Find the coordinates of the turning point.
- **c** Sketch the graph.

42 A parabola is described by the function $y = 2x^2 + 9x + 9$.

- **a** Find the x-intercepts of the parabola.
- **b** Find the y-intercept for this curve.
- **c** Find the axis of symmetry.
- **d** Find the y-coordinate of the vertex of the parabola.
- **e** Sketch the graph.

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Graphing quadratics using technology

43 Use your calculator to graph the equations below. Then answer the following questions:

i What is the vertex of the graph?

ii What is the y-intercept?

a $y = 4x^2 - 64x + 263$

b $y = -4x^2 - 48x - 140$

44 Use your calculator to graph $y = -3x^2 - 12$.

a What is the vertex of the graph?

b Are there any x-intercepts?

c For what values of x is the parabola decreasing?

45 Using a graphing calculator, sketch the curve $y = x^2 + 6.2x - 7$.

a Determine the axis of symmetry.

b Determine the minimum value of y.

46 Use a graphing calculator to sketch the parabola $y = -2x^2 + 16x - 24$.

a Find the x-intercepts of the parabola.

b Find the y-intercept of the parabola.

c Find the axis of symmetry of the parabola.

d Find the y-coordinate of the vertex of the parabola.

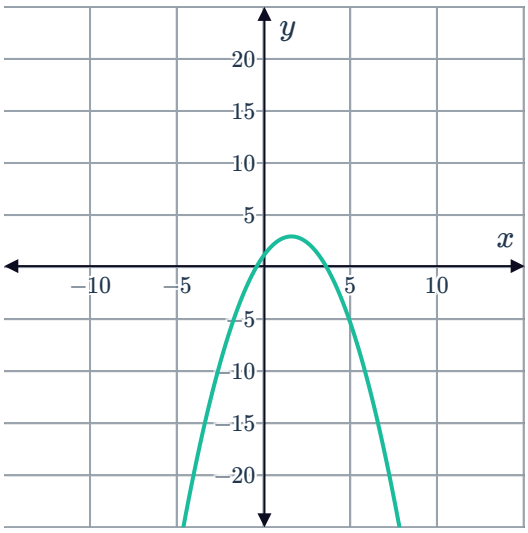
47 Consider the function $y = -0.72x^2 + \sqrt{5}x + 1.21$.

a Find the x-coordinate of the vertex to two decimal places.

b Find the y-coordinate of the vertex to two decimal places.

c Is the graph shown a possible viewing window on your calculator that shows the vertex and the x-intercepts?

d Use a graphing calculator to find the x-intercepts to two decimal places.



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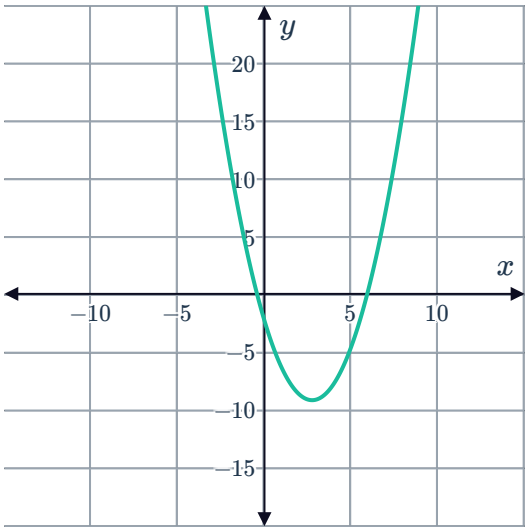
48 Consider the function $y = 0.91x^2 - 5x - \sqrt{5}$.

a Find the x-coordinate of the vertex to two decimal places.

b Find the y-coordinate of the vertex to two decimal places.

c Is the graph shown a possible viewing window on your calculator that shows the vertex and the x-intercepts?

d Use a graphing calculator to find the x-intercepts to two decimal places.



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